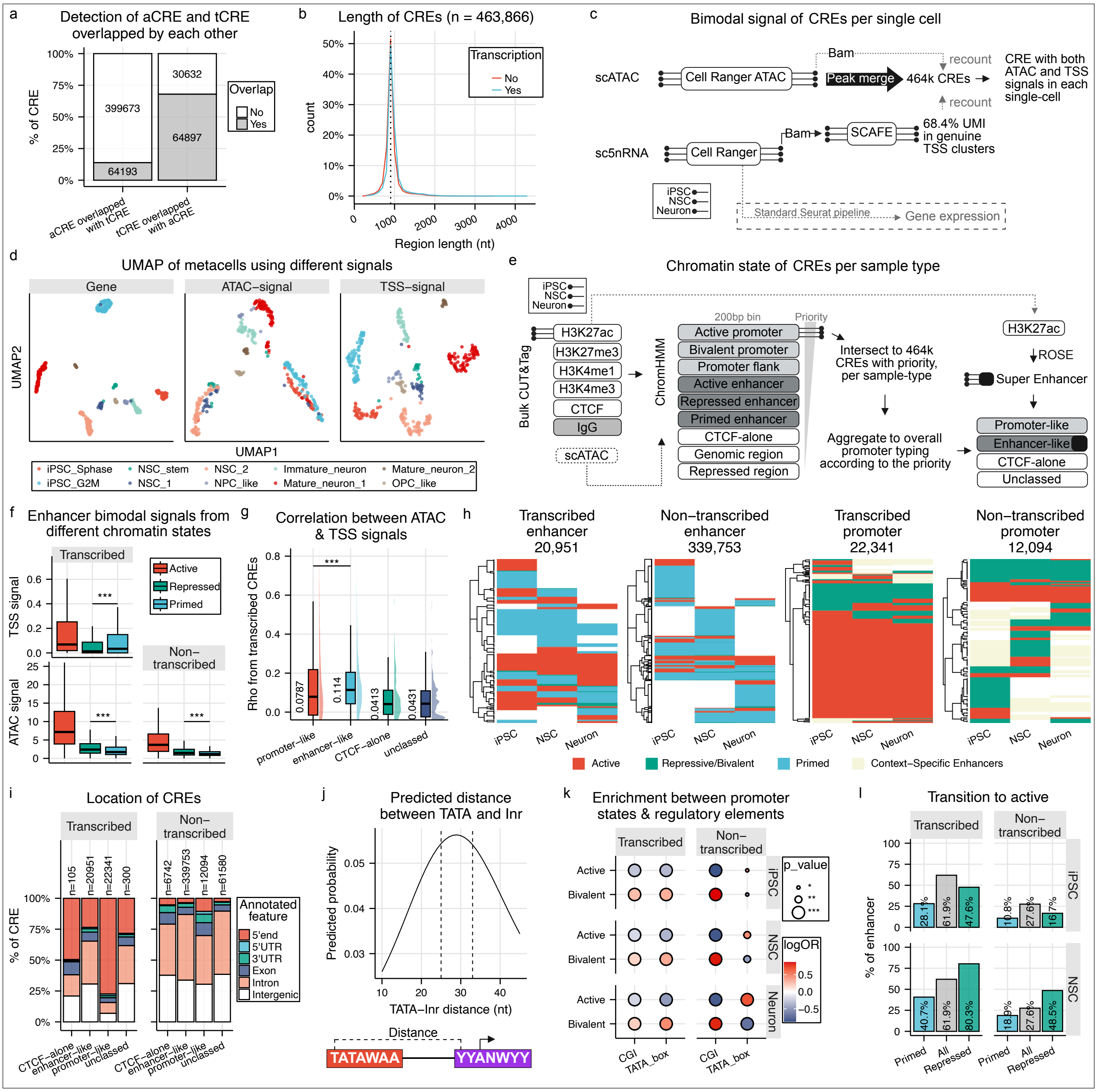

**Extended Figure 2 | Bimodal signals and genomic features of the defined CREs.**

**a**, Identification of aCRE derived from merged ATAC peaks and tCRE derived from SCAFE, and the percentage of them being overlapped with each other. **b**, Distribution of sequence length for all CREs, grouped as transcribed and non-transcribed. **c**, Schematic illustrating the quantification of bimodal signals at CREs using the scATAC-seq (accessibility) and sc5nRNA-seq (transcription start sites (TSS)) data. **d**, UMAP visualizations of metacells based on gene-signal, ATAC-signal and TSS-signal. **e**, Schematic showing the annotation of chromatin states of CREs based on bulk CUT&Tag from iPSC, NSC and Neuron using ChromHMM and the identification of super enhancer using ROSE. **f**, CRE Bimodal signals from different chromatin states, grouped in transcribed or non-transcribed and promoter or enhancer. Signals were aggregated from three samples. **g**, Spearman correlation between CRE ATAC and TSS signals across metacells in different classes of promoter-type. **h**, Heatmaps displaying the chromatin states transitions across the three samples. The CREs were stratified into transcribed enhancers, non-transcribed enhancers, transcribed promoters and non-transcribed promoters. **i**, Genomic distribution of CREs relative to annotated protein-coding gene coordinates. **j**, Predicted distance between TATA-box motif and Initiator motif from TATA-box mediated transcription sites (TSS tag >=3 inside the Initiator motif) with 25-33 nucleotides reaching 95% of maximum predicted probability. Schematics at the bottom show the positions where the distance measured. **k**, Enrichment between regulatory elements (CGI and TATA box) and chromatin states of promoter-like CREs. A subset of promoter-like CREs that are linked to annotated genes was used. **l**, Number and percentage of enhancer-like CREs with active chromatin state in at least one of the three samples across differentiation. Fisher's exact test was used to calculate odds ratios (OR). p < 0.05 (*), p < 0.01 (**), p < 0.001 (***), n.s. = not significant. For (**f**) and (**g**), statistical comparisons were performed using a two-sided Wilcoxon test.